Equilibrium Transition Dynamics in Heterogeneous-Agent Economies

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July 20, 2026

Abstract

Quantitative work with heterogeneous-agent models routinely computes perfect-foresight transition paths: an interest-rate path clears the asset market at every date while the wealth distribution evolves from given initial conditions. We ask when such equilibrium paths exist. In the continuous-time Huggett economy with CRRA utility, the operative equilibrium condition is market clearing in *flows*—aggregate consumption equals aggregate income plus interest on the bond supply—and the rate's only instantaneous leverage comes from the bond stock and from hand-to-mouth households, whose interest burden responds instantaneously to the rate. Our main theorem is a local fixed-point result built directly on this structure: because the hand-to-mouth channel transmits arbitrarily rapid oscillations of the rate path, the equilibrium map is *not* compact on the full band of continuous paths, and the fixed point must instead be found on an invariant class of paths sharing a common modulus of continuity, constructed from a temporal transmission estimate whose contemporaneous coefficient—the ratio of hand-to-mouth debt to bond supply—is strictly below one. Under that condition, and along transitions on which the borrowing constraint binds uniformly (the constrained population remains intact), equilibrium transitions exist over short horizons, and are unique (exactly under bounded wealth densities, or up to an exponentially small ambiguity in the realistic boundary-layer regime). We also show that whether households remain at the borrowing constraint near the end of a transition is governed by terminal conditions rather than by discounting, and we discuss, informally, why long horizons with risk aversion above one may admit multiple self-fulfilling transition paths.

JEL codes: C62, D52, E21, D31. *Keywords:* incomplete markets, borrowing constraints, transition dynamics, existence of equilibrium, mean field games.

1 Introduction

Modern quantitative macroeconomics runs on transition paths. To evaluate a credit crunch, a fiscal reform, or a monetary shock in a Bewley–Huggett–Aiyagari economy, one computes the perfect-foresight response: starting from an initial distribution of wealth, households optimize against a conjectured path of prices, the wealth distribution evolves under their decisions, and the price path is adjusted until markets clear at every date (Kaplan, Moll, and Violante, 2018; Boppart, Krusell, and Mitman, 2018; Auclert et al., 2021; Achdou et al., 2022a). The algorithms that do this are by now standard. The theory behind them is not: while existence and (under conditions) uniqueness of *stationary* equilibria are well understood (Aiyagari, 1994; Açıkgöz, 2018; Achdou et al., 2022a; Höfer, 2025), there is, to our knowledge, no theorem guaranteeing that the equilibrium transition path being computed exists.[1] This paper proves such a theorem for the canonical laboratory—the continuous-time Huggett economy with CRRA utility, two income states, a borrowing constraint, and a bond in fixed supply $B \geq 0$, over a horizon $[0, T]$—and shows that its shape is dictated by a specific economic mechanism.

The mechanism is easiest to see by asking: *who responds to today's interest rate today?* A household's consumption at date t is determined by its wealth and by the price path it expects from t onward; changing r at the single instant t leaves its date-t behavior unchanged. The wealth distribution at t depends on prices before t. So today's rate moves today's aggregate wealth not at all—market clearing stated in *stocks* is a degenerate restriction on the rate path (Proposition 4.3). The operative form of clearing is instead the *flow* form obtained by differentiating the stock condition along the distributional dynamics: clearing at every date is equivalent to clearing at the initial date—a restriction on the data—plus the aggregate budget identity

$$C(t) \;=\; Y(t) + r(t)B \qquad \text{at every } t,$$

aggregate consumption equals aggregate income plus interest on the bond supply (Lemma 4.2). This stock-to-flow conversion is already how the model is solved in practice: the transition algorithm of Achdou et al. (2022a) iterates on the rate path using the *time derivative* of excess bond supply (see the algorithm in their §5.5 and the accompanying Numerical Appendix and codes, Achdou et al., 2022b). What the flow form isolates analytically is the rate's contemporaneous leverage: differentiating the flow-clearing residual with respect to $r(t)$ gives exactly

$$-\big(B + |\underline{a}|\, m_1(t)\big),$$

where $m_1(t) := G_{1,t}(\{\underline{a}\})$ is the mass of hand-to-mouth households at the borrowing limit—

[1] The closest results are Ambrose (2021), on a related model of Achdou et al. (2014) with diffusive income, *no* borrowing constraint, and zero net supply, where small-time existence holds only for a *relaxed* clearing constraint and the exact problem is generically unsolvable; and Chen et al. (2025), an overlapping-generations economy with a never-binding natural borrowing limit and a small-lifespan restriction. Section 1.1 discusses both in detail.

the only agents whose date-t consumption depends on the date-t rate, through their budget $y_1 + r(t)\underline{a}$.

The main theorem. Our central result (Theorem 5.3 and Corollary 5.4) turns this structure into a short-horizon existence and uniqueness theorem for equilibrium transitions with $B > 0$. Two features of its architecture are dictated by the economics, and we want to flag them up front because they are where a naive approach fails.

First, *the equilibrium map is not compact on the full band of continuous rate paths*, so the textbook route—Schauder's fixed-point theorem applied to a map with relatively compact range—is unavailable. The reason is the hand-to-mouth channel itself: over short horizons the multiplier map Ψ_T (the rate that makes bond interest absorb excess consumption) tracks $A_0(r(t))$, where A_0 contains the term $m_1(0)\min\{c_{T,1}(\underline{a}), y_1 + r(t)\underline{a}\}/B$; whenever the hand-to-mouth branch is active, a rate path that ramps between two values over an arbitrarily short window produces an order-one oscillation of $\Psi_T(r)$ over that window (Proposition 5.1). Rapid oscillations of the rate pass straight through constrained households' budgets into aggregate consumption. No smoothing occurs on this channel, and no common modulus of continuity exists for the image of the full band.

Second, the fix is not technical patching but a sharper isolation of the economics. The map does satisfy a *temporal transmission estimate*: its output oscillates no more than $\bar{\theta}$ times the oscillation of the input plus a genuinely smoothing remainder,

$$|\Psi_T(r)(t) - \Psi_T(r)(s)| \;\le\; \bar{\theta}\,|r(t) - r(s)| + \omega_T(|t-s|), \qquad \bar{\theta} \;=\; \frac{|\underline{a}|\,\sup_t m_1(t)}{B} \;<\; 1,$$

where the coefficient $\bar{\theta}$ is the ratio of hand-to-mouth debt to the bond supply. When $\bar{\theta} < 1$, the class of paths whose modulus of continuity is $\Omega_T = \omega_T/(1-\bar{\theta})$—the fixed point of the modulus recursion $x \mapsto \bar{\theta}x + \omega_T$—is invariant under the map, and it *is* compact. Schauder's theorem applies there. Uniqueness holds by contraction when wealth densities are bounded near the constraint, and up to an ambiguity exponentially small in $1/T$ in the realistic regime where the density has the inverse-square-root boundary layer generated by the constraint. A further geometric point matters for the construction: the map is *antitone* in the rate (higher rates cut hand-to-mouth consumption), and an antitone map can throw an asymmetric band around its fixed point out of the band; invariance must be arranged on a *symmetric* interval around the anchoring rate r^{**} implied by the data (Remark 5.2).

The stability condition $\bar{\theta} < 1$ has a transparent reading, and it is, to our knowledge, new. A contemporaneous rise dr in the rate obliges the economy to absorb $B\,dr$ of additional interest income as consumption, while cutting hand-to-mouth consumption by $|\underline{a}|\,m_1\,dr$—the interest surcharge on constrained debt. The equilibrium feedback loop is stable when the second effect is smaller: hand-to-mouth debt below the outstanding stock of bonds. Economies with large constrained populations, deep borrowing, and thin asset markets are the ones in which equilibrium transitions are fragile.

Supporting results. Three further sets of results surround the main theorem.

The household problem along a rate path (Section 3, Appendices A–B). We provide the well-posedness package the main theorem consumes: unique constrained viscosity solutions of the household's Bellman equation along any admissible path, two-sided marginal-value bounds by explicit barriers, and a forward Kolmogorov theory for wealth distributions carrying a boundary atom driven by one-sided-Lipschitz flows. The economically substantive finding is that *whether the constraint binds near the end of the horizon is governed by terminal conditions, not by discounting*: with continuation values drawn from a stationary equilibrium at rate r^*, the constrained group stays put at the end of the transition if and only if $r(T) \geq r^*$, and disperses whenever the path ends below r^*—even if $r(t) < \rho$ throughout (Corollary 3.3). Constrained households are debtors; a terminal rate below the one embedded in their continuation plan lightens their debt service, and they save their way off the limit. Because $m_1(t)$ is also the market's contemporaneous lever, binding episodes and the tractability of the equilibrium problem switch on and off together.

Zero net supply (Section 6). With $B = 0$ the flow condition pins aggregate consumption to aggregate income and the *only* contemporaneous lever is the constrained mass. If the initial distribution carries an atom at the limit and the constraint remains binding, a short-horizon equilibrium exists (Proposition 6.1); if no one is constrained on a time interval, clearing there contains today's rate nowhere, no local solution theory applies, and—consistent with the nonexistence results of Ambrose (2021) in a frictionless relative of the model—exact clearing is generically the wrong demand; slightly elastic supply or a price rule is then the correct formulation, not an approximation (Proposition 6.2).

Long horizons (Section 7). What happens beyond short horizons is genuinely open, and we deliberately present it as informal discussion rather than as results: a monotonicity property of aggregate consumption in the rate path would localize all equilibria between two computable paths, but it is unproven even for intertemporal elasticity of substitution above one; and for risk aversion above one, economies constructed by Höfer (2025) with non-monotone stationary aggregate saving suggest singular linearizations of the clearing condition and multiple self-fulfilling transition paths.

1.1 Related literature

The stationary theory is by now well developed. Achdou et al. (2022a) formulate the continuous-time system, prove existence of stationary equilibria by the Aiyagari intermediate-value argument, and prove uniqueness when the intertemporal elasticity of substitution is at least one and borrowing is prohibited; Açıkgöz (2018) rigorizes the discrete-time counterpart and gives a calibrated multiplicity example. Mathematically complete stationary treatments are given by Shigeta (2023) (bounded utility) and Höfer (2025) (CRRA utility under a no-borrowing limit; ergodicity of the wealth process, continuity of aggregate assets in r, and the non-monotonicity construction for $\gamma > 1$ discussed in Section 7); Bayer, Rendall, and Wälde (2019) analyze the invariant wealth distribution at fixed prices. All of these concern steady states.

For transitions, the closest prior work is Ambrose (2021), who studies the household-wealth model of Achdou et al. (2014)—a variant with diffusive income, *no* borrowing constraint (compactly supported wealth densities), and zero net supply. He proves small-time existence and uniqueness for a *relaxed* problem in which exact clearing is replaced by constancy of the flow imbalance, and gives a nonexistence result (his §8) for the exact problem: with generic data, no exactly-clearing solution exists over short horizons. In our language, his model has $B = 0$ and, by construction, no mass at a borrowing constraint, so the rate's contemporaneous lever on clearing is identically zero; his Remark 1 notes explicitly that mass at a wealth boundary would contribute "another term proportional to r" through which clearing could be enforced. The present paper can be read as the constrained-economy counterpart: the point mass of hand-to-mouth households (and positive bond supply) restores exactly that missing term, and with it an existence theory for exact clearing. Chen et al. (2025) obtain a time-dependent market-clearing rate path in an overlapping-generations economy, under a never-binding natural borrowing limit and a short-lifespan restriction that plays the role of our short horizon.

Methodologically the paper sits in the mean field games (MFG) literature. Finite-horizon well-posedness results exist for price-mediated MFG in which the price is a multiplier on a constraint involving agents' *controls* (Gomes and Saúde, 2021; Fujii and Takahashi, 2022), for oligopoly models with prices that are explicit functionals of the distribution (Graber and Bensoussan, 2018), for MFG of controls (Cardaliaguet and Lehalle, 2018; Kobeissi, 2022), including a state-constrained case under a smallness condition on the dependence of preferences on the joint law of states and controls (Graber and Mayorga, 2021)—a condition that, in our multiplier formulation, scales like $1/B$ and fails precisely in the economically central small-B regime—and for general non-separable forward–backward parabolic systems over short horizons (Cirant, Gianni, and Mannucci, 2020). In all of these, either the clearing constraint responds nondegenerately to the contemporaneous price through the controls, or there is no exact clearing constraint at all; none features the combination present here—a hard borrowing constraint with a boundary point mass, switching income risk, a Hamiltonian singular at zero marginal value, and a price path pinned by instant-by-instant clearing. The deterministic theory of MFG with state constraints (Cannarsa and Capuani, 2018; Cannarsa, Capuani, and Cardaliaguet, 2021) supplies the right language for constrained trajectories; the planning problem (Porretta, 2014) treats a single terminal constraint enforced by a multiplier, where our clearing imposes one scalar constraint per instant. In discrete time, sequence-space methods (Auclert et al., 2021) work with linearizations of market-clearing conditions along price paths; our degeneracy result speaks to the conditioning of such Jacobians in the continuous-time limit, where stock-level clearing has vanishing diagonal while the flow form retains the invertible diagonal $B + |\underline{a}| m_1(t)$.

The paper proceeds as follows. Section 2 lays out the economy. Section 3 analyzes the household problem along a rate path. Section 4 develops the structure of market clearing. Section 5 contains the main theorem and its model corollary; Section 6 treats zero net supply. Section 7 discusses long horizons informally. Section 8 concludes. Appendices A–C contain the technical package and the verification of the model hypotheses.

2 The economy

2.1 Households

Time is continuous over a finite horizon $[0, T]$. There is a continuum of households with CRRA preferences

$$\mathbb{E}_0 \int_0^\infty e^{-\rho t}\, u(c_t)\, dt, \qquad u(c) = \frac{c^{1-\gamma}}{1-\gamma},\ \gamma > 0 \quad (u = \log \text{ for } \gamma = 1), \qquad \rho \geq 0.$$

A household's income y_t takes two values $0 < y_1 < y_2$ ("unemployed"/"employed"), switching with Poisson intensities $\lambda_1, \lambda_2 > 0$. Wealth is held in a riskless bond and evolves as

$$\dot{a}_t = y_t + r(t)\, a_t - c_t, \qquad a_t \ \geq\ \underline{a}, \qquad -\infty < \underline{a} < 0,$$

where $r(t)$ is the market interest rate, taken as given by the household, and $\underline{a}$ is an exogenous borrowing limit. The horizon is closed by a terminal valuation $v_{T,j}(a)$ of end-of-horizon wealth; the canonical choice, following Achdou et al. (2022a), is the value function $v_\infty(\cdot\,; r^*)$ of a stationary equilibrium with rate r^*, so that the finite-horizon problem is a truncation of a transition converging to a steady state.

The household's value functions $v_j(a, t)$, $j = 1, 2$, solve the backward Bellman (HJB) system

$$\rho v_j = \partial_t v_j + \max_{c \geq 0} \Big\{ u(c) + \partial_a v_j \left(y_j + r(t) a - c \right) \Big\} + \lambda_j \left(v_{-j} - v_j \right) \qquad \text{on } (\underline{a}, \infty) \times (0, T), \quad (1)$$

with terminal condition $v_j(\cdot, T) = v_{T,j}$ and the *state-constraint* boundary condition

$$\partial_a v_j(\underline{a}, t) \ \geq\ u'\big(y_j + r(t)\underline{a}\big), \qquad j = 1, 2, \qquad (2)$$

which says that the marginal value of wealth at the borrowing limit is at least the marginal utility of the consumption level $\underline{c}_j(t) := y_j + r(t)\underline{a}$ that just keeps the household at the limit; equivalently, that chosen saving at the limit is nonnegative, so the constraint is never violated. Optimal consumption satisfies $c_j = (u')^{-1}(\partial_a v_j)$ in the interior and $c_j(\underline{a}, t) = \min\{(u')^{-1}(\partial_a v_j(\underline{a}^+, t)),\, \underline{c}_j(t)\}$ at the limit; the saving policy is $s_j(a, t) = y_j + r(t)a - c_j(a, t)$.

The cross-sectional distribution of (income, wealth) is described by measures $G_j(\cdot, t)$ on $[\underline{a}, \infty)$ of total mass one, evolving under the households' saving policies and the income switching by the Kolmogorov forward equation, understood in a weak (measure) sense that accommodates a point mass

$$m_1(t) \ := \ G_{1,t}(\{\underline{a}\})$$

of households *at* the borrowing limit (Definition B.1 in Appendix B); $G_j(\cdot, 0) = G_{0,j}$ is given. This system—equations (12)–(17), together with the market-clearing condition (4), of Achdou et al. (2022a)—is the transition-dynamics version of the continuous-time Huggett model (Huggett, 1993).

2.2 Equilibrium

Definition 2.1. Given data (G_0, v_T) and bond supply $B \in [0, \infty)$, an *equilibrium transition* on $[0, T]$ is a continuous interest-rate path $r(\cdot)$, household value functions solving (1)–(2) with terminal data v_T (in the constrained viscosity sense, Definition A.3), and distributions solving the forward equation from G_0 under the induced policies, such that the bond market clears at every date:

$$S(t) \ := \ \sum_{j=1,2} \int_{[\underline{a},\infty)} a \, dG_j(a, t) \ = \ B \qquad \text{for all } t \in [0, T]. \tag{3}$$

Throughout, candidate rate paths lie in a band $\mathcal{K}_I = \{r \in C([0, T]) : r(t) \in I \ \forall t\}$, $I = [r_{\min}, r_{\max}]$, with

$$r_{\max} < \frac{y_1}{|\underline{a}|}, \tag{4}$$

which guarantees $\underline{c}_1(t) = y_1 + r(t)\underline{a} \geq \underline{c} > 0$: even the low-income household at the borrowing limit can service its debt out of income with something left to consume. Condition (4) is the transition version of the requirement that the borrowing limit be tighter than the natural limit; it is what makes the constraint economically meaningful (households at the limit are hand-to-mouth, not defaulting) and, mathematically, it is the inward-pointing condition on which the entire boundary analysis rests. The full list of maintained assumptions is collected in Appendix A (Assumptions A.1–A.5, covering primitives, the rate band, and the data); the terminal valuation is concave, increasing, with marginal values bounded above and decaying no faster than CRRA marginal utility of a linearly growing consumption level—properties satisfied by the canonical choice $v_\infty(\cdot\,; r^*)$.

For the quantitative equilibrium analysis we will distinguish two regimes for the wealth distribution near the constraint:

(D1) the absolutely continuous parts of the distributions have densities bounded near $\underline{a}$, uniformly in t and over the band;

(D1′) the densities may blow up like $(a - \underline{a})^{-1/2}$ near $\underline{a}$, with a constant uniform in t and over the band.

(D1′) is the realistic regime: it is exactly the profile generated by the square-root consumption law at a binding constraint (Section 3.1), just as in the stationary distribution of Achdou et al. (2022a). (D1) holds, for instance, transiently while the constraint does not bind.

2.3 Why transitions are not steady states strung together

In a stationary equilibrium the logic is: fix r, solve the household problem, compute the implied stationary wealth stock $S_{\text{stat}}(r)$, and find r with $S_{\text{stat}}(r) = B$. The map $r \mapsto S_{\text{stat}}(r)$ is a well-behaved scalar function, and existence follows from its boundary behavior (Aiyagari, 1994; Achdou et al., 2022a). One might hope to treat a transition as a family of such problems indexed by t. This fails, for an economic reason: $S(t)$ is a *stock*, accumulated from past saving decisions, each of which was made looking at *future* rates. The rate at date t influences wealth at date t not at all; it influences wealth after t through household budgets, and consumption before t through anticipation. Section 4 makes this precise and locates the two places where the contemporaneous rate does bite: interest on the bond supply, and the budgets of constrained households. The main theorem of Section 5 is built entirely on those two levers; and the second lever is itself an equilibrium object, switched on and off by the binding analysis of the next section.

3 Household behavior along an interest-rate path

This section takes the rate path $r \in \mathcal{K}_I$ as given and establishes what we need about individual behavior: that the household problem is well posed (Theorem 3.1), and—the economically substantive part—when households at the borrowing limit stay there (Section 3.1). Readers willing to take well-posedness on faith can skim to Section 3.1; the technical package, including the regularity and stability estimates used later, is developed in Appendix A.

Theorem 3.1 (the household problem is well posed)**.** *Under the maintained assumptions (Appendix A), for every $r \in \mathcal{K}_I$ the household's value function is the unique constrained viscosity solution of (1)–(2) with terminal data v_T, in the class of continuous functions concave and increasing in wealth with linear growth. It is continuously differentiable in wealth on $(\underline{a}, \infty)$, with a one-sided derivative at $\underline{a}$; consumption is continuous and nondecreasing in wealth; marginal values admit two-sided bounds, uniform over the band,*

$$0 < \kappa\, e^{-CT}(1 + a - \underline{a})^{-\gamma} \;\leq\; \partial_a v_j(a,t) \;\leq\; \max\{\bar{p}_T,\, u'(\underline{c})\}\, e^{(r_{\max}-\rho)^{+}T},$$

where $\bar{p}_T$ bounds the terminal marginal value; equivalently, consumption is bounded below by a positive constant and grows at most linearly in wealth. The value function and the policies *depend continuously—and the value Lipschitz-continuously—on the rate path, with only the* future *of the path entering (Lemma A.9).*

Three comments on the economics of the proof (details in Appendix A). First, the bounds on marginal value are where finite horizon differs from the steady state. Stationarily, the binding constraint supplies the boundary marginal value *exactly*—$\partial_a v_1(\underline{a}) = u'(y_1 + r\underline{a})$ whenever $r < \rho$—and concavity does the rest. Along a transition, binding is not guaranteed (next subsection), and the bounds must be built by comparison with explicit sub- and super-solutions: the upper bound propagates backward the larger of the terminal marginal value and the boundary obstacle $u'(\underline{c})$; the lower bound is a CRRA-shaped barrier expressing that consumption cannot outgrow wealth linearly. The cap $\bar{p}_T$ on terminal marginal values is a genuine assumption on the terminal data, not a free good. Second, the bounds keep marginal values in a compact positive range, which is what tames the CRRA Hamiltonian $H(p) = \max_c\{u(c) - pc\}$, whose slope $H'(p) = -p^{-1/\gamma}$ blows up as $p \downarrow 0$. Third—an honest simplification—neither $\rho > 0$ nor $r < \rho$ appears anywhere: on a finite horizon the terminal valuation replaces transversality, and impatience plays no coercive role. The conditions involving ρ reappear only through the terminal data, which is exactly where the next subsection locates them.

3.1 When does the borrowing constraint bind?

Say the constraint *binds* at date t if households at the limit stay there: $s_1(\underline{a}, t) = 0$, equivalently $\partial_a v_1(\underline{a}^+, t) = u'(\underline{c}_1(t))$. When it binds, the point mass $m_1(t)$ of hand-to-mouth households persists and grows; when it fails ($s_1(\underline{a}, t) > 0$), the constrained group saves its way off the limit and the mass disperses. In the stationary economy binding is automatic under impatience, $r < \rho$. Along a transition the governing comparison is different:

Lemma 3.2 (binding near the end of the horizon)**.** *(i) If the terminal marginal value at the limit exceeds the marginal utility of hand-to-mouth consumption at the terminal rate—$\partial_a v_{T,1}(\underline{a}) > u'(y_1 + r(T)\underline{a})$—then the constraint does* not *bind on an interval $(T - \delta, T)$, regardless of how $r(t)$ compares to ρ. (ii) If the reverse strict inequality holds, the constraint binds near T.*

Corollary 3.3 (the knife edge at the canonical terminal data)**.** *Let $v_T = v_\infty(\cdot\,; r^*)$, the stationary value at rate r^*, so that $\partial_a v_{T,1}(\underline{a}) = u'(y_1 + r^*\underline{a})$ by the stationary binding equality. Then the boundary saving rate at the end of the horizon is*

$$s_1(\underline{a}, T^-) \;=\; \big[(y_1 + r(T)\underline{a}) - (y_1 + r^*\underline{a})\big]^+ = \big[(r(T) - r^*)\,\underline{a}\big]^+,$$

and since $\underline{a} < 0$: the constraint binds at T^- if and only if $r(T) \geq r^*$. *Whenever the path ends below the rate embedded in the continuation values, the hand-to-mouth group disperses near the end of the transition—even if $r(t) < \rho$ throughout.*

The economics bears repeating because it inverts a steady-state intuition. Households at the borrowing limit are *debtors*: their consumption is income net of debt service, $y_1 + r(t)\underline{a}$, which *falls* as the rate rises. Their continuation plan, embedded in v_T, was drawn at rate r^*. If the transition delivers a terminal rate below r^*, debt service is lighter than the continuation plan assumed; the constrained household finds itself with slack, saves, and detaches from the limit. Impatience never enters: over a short remaining horizon, behavior is governed by the terminal valuation, not by discounting. A closed-form illustration with no income risk, in which binding fails on an entire terminal interval despite $r < \rho$, is given in Example A.8 (Appendix A); the same example exhibits the turnpike property—binding is restored at dates far from T, where the stationary logic reasserts itself.

Near the constraint, consumption follows a square-root law, as in the stationary analysis of Achdou et al. (2022a), but with a time-dependent speed:

Lemma 3.4 (square-root law along a transition)**.** *Let $r \in C^1$ and suppose the constraint binds on $[t_0, t_1] \subset (0, T)$. Define*

$$\nu_1(t) \;:=\; \frac{(\rho + \lambda_1 - r(t))\, u'(\underline{c}_1(t)) - \lambda_1\, \partial_a v_2(\underline{a}, t)}{-u''(\underline{c}_1(t))} \;+\; \dot{r}(t)\, \underline{a}.$$

If $\nu_1 \geq \underline{\nu} > 0$ on $[t_0, t_1]$, then locally uniformly there

$$c_1(a,t) = \underline{c}_1(t) + \sqrt{2\nu_1(t)}\,\sqrt{a - \underline{a}} + O(a - \underline{a}), \qquad s_1(a,t) \sim -\sqrt{2\nu_1(t)}\,\sqrt{a - \underline{a}}.$$

Relative to the stationary coefficient, time dependence adds the term $\dot{r}(t)\,\underline{a}$: since $\underline{a} < 0$, a *rising* rate path lowers ν_1—anticipated increases in debt service slow the approach to the constraint—and $\nu_1(t) \downarrow 0$ is precisely the incipient dispersal regime of Corollary 3.3. We stress the hypotheses: a C^1 path, binding on the interval, and ν_1 bounded away from zero. The square-root layer is a property of *binding regimes*, not of the model wholesale; this matters for the hypotheses of the main theorem, and we return to it in Section 5.3.

Finally, the constrained mass itself. While the constraint binds, $m_1(t)$ grows by the inflow of low-income households hitting the limit and shrinks only as its members draw the high income state (at rate λ_1). When binding fails, the entire mass detaches *at once* and moves into the interior as a traveling atom: $m_1(t)$ drops to zero discontinuously (Proposition B.4, Appendix B).

4 Market clearing along a transition

We now turn to equilibrium. Throughout, $S(t)$ denotes aggregate wealth (3), $C(t) = \sum_j \int c_j \, dG_{j,t}$ aggregate consumption, and $Y(t) = \sum_j \int y_j \, dG_{j,t}$ aggregate income.

Proposition 4.1 (the data must clear the market at date zero)**.** *If an equilibrium transition exists, then* $S(0) = \sum_j \int a\, dG_{0,j} = B$*. No choice of prices can repair incompatible initial data: clearing at* $t = 0$ *involves the initial distribution alone.*

This is obvious once stated, but it is a substantive discipline on policy experiments: an "MIT shock" that revalues wealth or bonds at $t = 0$ must do so consistently with (3), or no equilibrium path exists at all.

Lemma 4.2 (clearing in stocks equals clearing at date zero plus clearing in flows)**.** *Along any solution of the distributional dynamics with uniformly bounded first moments,* $t \mapsto S(t)$ *is Lipschitz and*

$$\dot{S}(t) \;=\; \sum_j \int s_j\, dG_{j,t} \;=\; Y(t) + r(t)S(t) - C(t),$$

the boundary contributing nothing (households at the limit have zero saving while the constraint binds; dispersing mass is counted in the interior). Consequently, for $r \in \mathcal{K}_I$*:*

$$S(t) = B \;\; \forall t \in [0,T] \qquad \Longleftrightarrow \qquad S(0) = B \;\; \textit{and} \;\; C(t) = Y(t) + r(t)B \;\; \forall t \in [0,T].$$

The flow form is the economy's aggregate budget identity: consumption absorbs income plus the interest paid on the bond supply. It is also the form in which the model is solved numerically: the transition algorithm of Achdou et al. (2022a, §5.5) updates the rate path by the time derivative of excess bond supply, $r^{\ell+1}(t) = r^{\ell}(t) - \xi(t)\frac{d}{dt}S^{\ell}(t)$, precisely the stock-to-flow conversion above; see their Numerical Appendix and codes (Achdou et al., 2022b). The reason the flow form is the right object for analysis as well is the following degeneracy of the stock form.

Proposition 4.3 (stock-level clearing has no contemporaneous diagonal)**.** *Fix data and a path* $r \in \mathcal{K}_I$*, and consider the map* $r \mapsto S(\cdot\,; r)$*. Then: (i)* $S(0; r)$ *is independent of* r*; (ii) for path perturbations* δr *supported on a window* $(t_0 - h, t_0 + h)$*,*

$$|S(t_0; r + \delta r) - S(t_0; r)| \;\leq\; C\, h\, (1 + T)\, \|\delta r\|_\infty$$

under (D1), with an additional factor $1 + \log_+\big(1/(h\|\delta r\|_\infty)\big)$ *under (D1′)—either way the bound vanishes with* h*: the rate at date* t_0 *has no contemporaneous effect on wealth at date* t_0*; and (iii) the linearized map* $DS(\cdot\,; r)$ *is a compact integral operator on* $C([0,T])$*, so the equation* $S(\cdot\,; r) = B$ *is a first-kind equation admitting no bounded local inverse.*

By contrast, differentiating the flow-clearing residual with respect to the contemporaneous rate,

$$\frac{\partial}{\partial r(t)}\Big[C(t) - Y(t) - r(t)B\Big] \;=\; m_1(t)\, \underline{a} \;-\; B \;=\; -\big(B + |\underline{a}|\, m_1(t)\big), \tag{5}$$

because the only households whose date-t consumption depends on the date-t rate are those at the borrowing limit, consuming $y_1 + r(t)\underline{a}$ (only the low-income type carries boundary mass, by Assumption A.6). The flow constraint has a uniformly invertible diagonal if and only if $B + |\underline{a}|m_1(t)$ is bounded away from zero—automatically for $B > 0$; through the constrained mass, hence through the binding analysis of Section 3.1, when $B = 0$.

5 The main theorem: short-horizon existence and uniqueness

Throughout this section $B > 0$ and $S(0) = B$ (Proposition 4.1). By Lemma 4.2, equilibrium is equivalent to a fixed point over rate paths: $r = \Psi_T(r)$ with

$$\Psi_T(r)(t) \;:=\; \frac{C(t;r) - Y(t)}{B},$$

the rate that makes bond interest absorb the excess of consumption over income implied by the path r. (Aggregate income $Y(t)$ does not depend on the path: the income-type marginals follow the autonomous two-state chain.)

5.1 Why the fixed point must be local

Over short horizons the multiplier map tracks an *anchoring function* of the contemporaneous rate. Let $c_{T,j} := (u')^{-1}(\partial_a v_{T,j})$ denote the terminal-implied consumption policies and define

$$A_0(x) \;:=\; \frac{1}{B}\Big[\textstyle\sum_j \int_{(\underline{a},\infty)} c_{T,j}\, dG_{0,j} + m_1(0)\, \min\{c_{T,1}(\underline{a}),\, y_1 + x\,\underline{a}\} - Y(0)\Big], \qquad x \in I. \quad (6)$$

A_0 is nonincreasing and Lipschitz with constant $\theta_0 := m_1(0)|\underline{a}|/B$: its only dependence on the rate is the budget of the initially constrained households. (Only the type-1 atom enters: the high-income type is unconstrained at the limit throughout, by Assumption A.6.) The *anchoring approximation*, over a band $\mathcal{K}_J$ of paths with values in a subinterval $J \subseteq I$, is the statement

$$\sup_{r\in\mathcal{K}_J}\,\sup_{t\in[0,T]} \big|\Psi_T(r)(t) - A_0(r(t))\big| \;\le\; \varepsilon_T, \qquad \varepsilon_T \to 0 \;\text{ as } T \to 0. \quad (7)$$

Appendix C establishes (7) on the symmetric sub-band used by the main theorem, on uniform binding regimes, as hypothesis (M1) of Corollary 5.4.

The approximation (7) immediately reveals that the textbook compactness route is closed:

Proposition 5.1 (no compactness)**.** *Suppose* (7) *holds on* $\mathcal{K}_J$ *and* A_0 *is nonconstant on* J*, with* $2\varepsilon_T < \operatorname{osc}_J A_0 := \sup_{x,y\in J}|A_0(x) - A_0(y)|$*. Then* $\Psi_T(\mathcal{K}_J)$ *is not uniformly equicontinuous, hence not relatively compact in* $C([0,T])$*.*

Proof. Choose $x_0, x_1 \in J$ with $d := |A_0(x_0) - A_0(x_1)| > 2\varepsilon_T$. For any $h > 0$ pick $s_h < t_h$ in $[0,T]$ with $t_h - s_h < h$ and a continuous path $r_h \in \mathcal{K}_J$ with $r_h(s_h) = x_0$, $r_h(t_h) = x_1$ (a linear ramp on $[s_h, t_h]$ suffices). By (7), $|\Psi_T(r_h)(t_h) - \Psi_T(r_h)(s_h)| \geq d - 2\varepsilon_T > 0$, a lower bound independent of h: no common modulus of continuity exists, and Arzelà–Ascoli rules out relative compactness. □

A_0 is nonconstant precisely when $m_1(0) > 0$ and the budget branch of the min in (6) is active on a nondegenerate subinterval—that is, when the hand-to-mouth diagonal is operative. The economics: rapid oscillations of the rate path pass *undamped, with coefficient* θ_0, through constrained households' budgets into the multiplier map; the map performs no smoothing on this channel. Any fixed-point argument must therefore work on a restricted class of paths whose oscillations are controlled—and the restriction must be self-consistent, i.e. invariant under the map.

Remark 5.2 (antitone geometry). A second, more elementary point shapes the construction. A_0 is *antitone*: higher rates cut hand-to-mouth consumption. An antitone map can throw an asymmetric band around its fixed point out of the band: on $I = [0, 10]$, the map $A_0(x) = \frac{3}{2} - \frac{1}{2}x$ is $\frac{1}{2}$-Lipschitz with interior fixed point $x^{**} = 1$, yet $A_0(10) = -\frac{7}{2} \notin I$. Deviations *below* the fixed point are produced by rates *above* it. Invariance must therefore be arranged on a *symmetric* interval $[r^{**} - R,\ r^{**} + R]$ around the anchoring rate, on which a q-Lipschitz antitone map with $q < 1$ is genuinely contracting toward its fixed point.

5.2 The fixed-point theorem

The following theorem is the paper's central mathematical result. It is stated abstractly—Ψ_T any family of maps on rate paths—because its hypotheses are exactly the economic estimates the model must deliver, and stating them separately makes the division of labor transparent. A *modulus* is a nondecreasing function $\omega : [0, T] \to [0, \infty)$ with $\omega(0) = 0$ and $\omega(h) \to 0$ as $h \downarrow 0$.

Theorem 5.3 (short-horizon fixed point)**.** *Let $I = [r_{\min}, r_{\max}]$ and let $A_0 : I \to \mathbb{R}$ be Lipschitz with constant $q < 1$ and fixed point $r^{**} \in \operatorname{int} I$. Choose $R > 0$ with $I_R := [r^{**}-R,\ r^{**}+R] \subset \operatorname{int} I$ and set $\mathcal{K}_R := \mathcal{K}_{I_R}$. For each sufficiently small $T > 0$ let $\Psi_T : \mathcal{K}_R \to C([0, T])$ satisfy:*

(H1) Anchoring approximation: *there is $\varepsilon_T \geq 0$ with*

$$\sup_{r \in \mathcal{K}_R} \sup_{t \in [0,T]} |\Psi_T(r)(t) - A_0(r(t))| \leq \varepsilon_T, \qquad \varepsilon_T < (1 - q)\, R.$$

(H2) Continuity: $\Psi_T : (\mathcal{K}_R, \|\cdot\|_\infty) \to C([0, T])$ *is continuous.*

(H3) Temporal transmission estimate: *there are $\bar{\theta} < 1$ and a modulus ω_T such that, for every $r \in \mathcal{K}_R$ and all $s, t \in [0, T]$,*

$$|\Psi_T(r)(t) - \Psi_T(r)(s)| \ \leq\ \bar{\theta}\, |r(t) - r(s)| + \omega_T(|t - s|). \tag{8}$$

Then Ψ_T has a fixed point in $\mathcal{K}_R$.

If, in addition, there is $C_3 \geq 0$ with

$$\|\Psi_T(r) - \Psi_T(r')\|_\infty \leq (\bar{\theta} + C_3 T)\,\|r - r'\|_\infty \qquad (r, r' \in \mathcal{K}_R), \tag{9}$$

then, whenever $\bar{\theta} + C_3 T < 1$, the fixed point is unique in $\mathcal{K}_R$ and Picard iteration converges geometrically from every starting path in $\mathcal{K}_R$.

Finally, set $D := 2R$ and suppose instead of (9) that for some $C_1, C_2 \geq 0$, all distinct $r, r' \in \mathcal{K}_R$ with $\eta = \|r - r'\|_\infty \in (0, D]$ satisfy

$$\|\Psi_T(r) - \Psi_T(r')\|_\infty \;\leq\; \bar{\theta}\,\eta + C_1 T\,\eta + C_2 T\,\eta\Big(1 + \log\frac{D}{\eta}\Big). \tag{10}$$

Then any two fixed points in $\mathcal{K}_R$ satisfy

$$\|r - r'\|_\infty \;\leq\; D\,\exp\Big(-\frac{1 - \bar{\theta} - (C_1 + C_2)T}{C_2 T}\Big) \tag{11}$$

whenever $C_2 > 0$ and the numerator is positive; if $C_2 = 0$ and $\bar{\theta} + C_1 T < 1$, the fixed point is unique.

Proof. Step 1: invariance of the local band. For $r \in \mathcal{K}_R$ and any t, using (H1), $A_0(r^{**}) = r^{**}$, and the q-Lipschitz property,

$$|\Psi_T(r)(t) - r^{**}| \leq |\Psi_T(r)(t) - A_0(r(t))| + |A_0(r(t)) - A_0(r^{**})| \leq \varepsilon_T + q\,|r(t) - r^{**}| \leq \varepsilon_T + qR < R.$$

Thus $\Psi_T(\mathcal{K}_R) \subset \mathcal{K}_R$.

Step 2: a compact invariant class. Define $\Omega_T(h) := \omega_T(h)/(1 - \bar{\theta})$ and

$$X_T := \big\{r \in \mathcal{K}_R : \; |r(t) - r(s)| \leq \Omega_T(|t - s|) \;\; \forall s, t \in [0, T]\big\}.$$

X_T is nonempty (it contains the constant path r^{**}), convex, closed in the uniform norm, uniformly bounded, and uniformly equicontinuous—hence compact, by Arzelà–Ascoli. If $r \in X_T$, then (H3) gives

$$|\Psi_T(r)(t) - \Psi_T(r)(s)| \leq \bar{\theta}\,\Omega_T(|t - s|) + \omega_T(|t - s|) = \Omega_T(|t - s|),$$

since Ω_T is precisely the fixed point of the modulus recursion $x \mapsto \bar{\theta}x + \omega_T$. Together with Step 1 this shows $\Psi_T(X_T) \subset X_T$. By (H2), $\Psi_T|_{X_T}$ is continuous; Schauder's theorem gives a fixed point in $X_T \subset \mathcal{K}_R$.

Step 3: contraction. Under (9), Ψ_T is a strict contraction of the complete metric space $\mathcal{K}_R$ (closed in $C([0, T])$; self-map by Step 1) whenever $\bar{\theta} + C_3 T < 1$; Banach's theorem gives uniqueness and geometric Picard convergence.

Step 4: the ambiguity estimate. Let $r \neq r'$ be fixed points and $\eta = \|r - r'\|_\infty$; both paths take values in I_R, so $0 < \eta \leq D$. Applying (10) to the fixed-point identity and dividing by η,

$$1 \;\leq\; \bar{\theta} + (C_1 + C_2)T + C_2 T \,\log \frac{D}{\eta}.$$

If $C_2 > 0$, rearranging gives (11); if $C_2 = 0$, the inequality contradicts $\bar{\theta} + C_1 T < 1$, so two distinct fixed points cannot exist. □

The theorem's architecture mirrors the economics. Step 1 is stabilization of the *level* of rates by the anchoring data, on the symmetric interval that the antitone geometry requires (Remark 5.2). Step 2 handles the *oscillation*: (H3) says the map transmits temporal oscillations of the rate with gain $\bar{\theta}$—the hand-to-mouth exposure—plus a smoothing remainder; when $\bar{\theta} < 1$ the transmitted oscillation contracts, and the class of paths whose modulus is the recursion's fixed point Ω_T is exactly the self-consistent restriction that Proposition 5.1 demands. The two uniqueness regimes correspond to the two density regimes at the constraint.

5.3 The model corollary

Corollary 5.4 (equilibrium transitions exist for short horizons)**.** *Let $B > 0$, $S(0) = B$, and let the anchoring function A_0 of* (6) *have Lipschitz constant $\theta_0 \leq \bar{\theta} < 1$ and fixed point $r^{**} \in \operatorname{int} I$; fix a symmetric sub-band $I_R \subset \operatorname{int} I$ as in Theorem 5.3. Suppose the model delivers, on $\mathcal{K}_R$ and for small T:*

- (M1) *the anchoring approximation* (7) *with $\varepsilon_T < (1 - \bar{\theta})R$;*
- (M2) *continuity of $r \mapsto C(\cdot\,; r)$ in the uniform norm;*
- (M3) *the temporal transmission estimate* (8) *with coefficient $\bar{\theta}$ bounded by the hand-to-mouth exposure $|\underline{a}| \sup_{t,r} m_1(t; r)/B$;*
- (M4) *either the Lipschitz estimate* (9) *(bounded densities, (D1)) or the log-Lipschitz estimate* (10) *(boundary layer, (D1$'$)).*

Then there is $T^* > 0$, depending on the data, $\bar{\theta}$, and the sub-band, such that for all $T \leq T^*$ an equilibrium transition exists with rate path in $\mathcal{K}_R$; it is unique in $\mathcal{K}_R$ under (M4)-(D1), and unique up to the exponentially small ambiguity (11) *under (M4)-(D1$'$). (A fixed point of Ψ_T satisfies $C = Y + rB$ at every date; with $S(0) = B$, Lemma 4.2 upgrades this to $S \equiv B$.)*

The corollary's hypotheses (M1)–(M4) are the precise interface between the equilibrium theorem and the household-level analysis, and we state them as hypotheses deliberately: they are aggregate estimates, and what the household package of Appendices A–B delivers depends on the regime. Appendix C gives the verification arguments and their exact status. In brief:

- The *contemporaneous ingredients* are exact and unconditional: constrained households consume $y_1 + r(t)\underline{a}$, so the atom's contribution to $C(t)$ responds to the rate with coefficient $m_1(t)\underline{a}$—this is what puts the hand-to-mouth exposure into (M3) and the θ_0-Lipschitz constant into A_0.

- The *smoothing ingredients*—the modulus ω_T in (M3), the remainder ε_T in (M1), and the interior parts of (M2) and (M4)—are delivered by the household package on *uniform binding regimes*: when the constraint binds along the sub-band (sufficient conditions: the terminal compatibility $r_{\min} \geq r^*$ of Corollary 3.3 for canonical terminal data, plus paths slowly varying enough that the square-root coefficient ν_1 of Lemma 3.4 stays positive), the atom mass is absolutely continuous with bounded derivative, the boundary layer has the square-root profile, and the interior policies have the stability in the path recorded in Lemma A.9 and Remark A.10.

- One strengthening of the data assumptions is genuinely needed, and we record it as Assumption A.4 (Appendix A): the maintained bounds on terminal marginal values do *not* by themselves imply a square-root modulus for the terminal consumption policy near the limit—terminal data with $c_T(a) = c_0 + (a - \underline{a})^{\alpha}$, $\alpha < \frac{1}{2}$, satisfy every maintained bound yet have a worse modulus (Remark A.1). Since the layer estimates propagate backward from the terminal data, (M4)-(D1′) requires the terminal policy itself to have the square-root modulus; canonical terminal data $v_\infty(\cdot\,; r^*)$ satisfy this automatically, by the stationary square-root law.

Economically, the corollary says: *short equilibrium transitions exist—and the natural computational scheme converges to them under bounded densities, and to within the exponentially small ambiguity in the boundary-layer regime—in economies where hand-to-mouth debt is smaller than the bond supply and along transitions that keep the constrained population intact.* Both qualifiers are substantive. The first is the stability condition $\bar{\theta} < 1$ discussed in the introduction. The second connects the equilibrium theory back to Section 3.1: dispersal episodes—terminal rates below the continuation rate, sharp rate rises—switch off both the square-root layer that (M4) quantifies and, at $B = 0$, the market's contemporaneous lever itself.

6 Zero net supply

At $B = 0$ the flow condition reads $C(t; r) = Y(t)$—aggregate consumption pinned to aggregate income—and by (5) the only contemporaneous lever of the rate on clearing is the constrained mass, with weight $|\underline{a}|\, m_1(t)$. The fixed-point formulation solves the flow condition for the rate through the constrained households' budget:

$$r(t) \;=\; \widetilde{\Psi}_0(r)(t) \;:=\; \frac{C^{\text{int}}(t; r) + m_1(t; r)\, y_1 - Y(t)}{|\underline{a}|\, m_1(t; r)},$$

with C^{int} the consumption of the unconstrained.

Proposition 6.1 ($B = 0$ with a constrained population)**.** *Let $B = 0$, $S(0) = 0$, $m_1(0) > 0$, and assume—as an explicit hypothesis—that the constraint binds along the sub-band throughout $[0, T]$ (necessary for this is the terminal-data compatibility $\partial_a v_{T,1}(\underline{a}) \le u'(y_1 + r_{\min}\underline{a})$, i.e. $r_{\min} \ge r^*$ for canonical terminal data; sufficient, in addition, is a restriction to slowly varying paths keeping $\nu_1 > 0$). Then the constrained mass decays no faster than its members find high income, $m_1(t) \ge m_1(0)e^{-\lambda_1 t}$, the map $\widetilde{\Psi}_0$ is well defined with uniformly positive denominator, and—under the analogues of (M1)–(M4) for $\widetilde{\Psi}_0$ on a symmetric sub-band around its anchoring rate—an equilibrium transition exists for $T \le T^*(\text{data}, m_1(0))$, unique under (D1). The horizon T^* shrinks linearly with $m_1(0)$: a thin hand-to-mouth population supports only a thin theorem.*

Proposition 6.2 ($B = 0$ with no constrained population: no local theory)**.** *Suppose $B = 0$ and $m_1 \equiv 0$ on some interval—for instance near $t = 0$ when the initial distribution has no mass at the limit, or following a dispersal episode. On that interval the clearing condition $C(t; r) = Y(t)$ contains the contemporaneous rate* nowhere*: its linearization in the path is a compact operator (as in Proposition 4.3), the constraint is a first-kind equation, and no local solution theory—Picard, Newton, or implicit function—applies. In this regime relaxations are a necessity, not a computational convenience: an ε-elastic bond supply $B_\varepsilon(r) = \varepsilon r$, or a price rule $r(t) = F(C(t) - Y(t))$ with F Lipschitz, restores an invertible diagonal $\varepsilon + |\underline{a}|m_1$ and places the problem back in the scope of Theorem 5.3; as $\varepsilon \to 0$ one retains only weak limits of approximate equilibria, with clearing recovered in an integrated sense. Existence of exact equilibria in this regime is open—and the neighbouring analysis of Ambrose (2021), in a model where the constrained-mass channel is absent by construction, shows that there the exact problem is generically* unsolvable *while its relaxation is well posed over short horizons.*

The pair delivers a sharp message about the textbook case $B = 0$: *the hand-to-mouth population is the market.* When it is present, its budget is the margin along which the interest rate clears the market instant by instant. When it is absent, the interest rate is a shadow price with no contemporaneous market to clear, exact instant-by-instant clearing overdetermines the system, and the economically meaningful formulation gives the asset supply some elasticity or the price some friction.

7 Long transitions: an informal discussion

The results above are local in the horizon, and we want to be candid that what follows is not a set of theorems but a map of the open territory. The anticipation and history channels of the multiplier map each carry a factor T, so the "loop gain" of the expectations feedback—beliefs about future rates alter consumption today, which alters the wealth distribution, which alters market-clearing rates tomorrow—scales like T^2; beyond the horizon T^* of Corollary 5.4, contraction-based uniqueness must fail *as an argument*, and something else is needed.

The natural candidate, suggested by the stationary theory, is monotonicity in the rate. It is worth recording first why this is not a soft comparison-principle fact. Couple two economies under rate paths $r_h \geq r_\ell$ (pointwise), with identical income realizations and initial wealth. Wealth under the higher path dominates only if, wherever wealth is negative, the consumption cut induced by higher rates exceeds the added interest burden: $c^{r_\ell}(a,t) - c^{r_h}(a,t) \geq (r_h(t) - r_\ell(t))\,|a|$. For debtors this is a quantitative restriction on the consumption response, not a sign restriction. In the stationary economy, Achdou et al. (2022a) prove the aggregate version—saving increasing in r—when the intertemporal elasticity of substitution is at least one ($\gamma \leq 1$) and borrowing is prohibited, and conclude uniqueness of the steady state. Their argument does not survive transplantation to time-varying paths, for identifiable reasons: it perturbs by a *constant* rate (a time-varying perturbation turns the linearized Euler system into a two-point boundary problem with sign-indefinite kernels mixing substitution and wealth effects); it uses CRRA homogeneity to absorb the rate shift (which requires terminal data covariant under the same scaling—and terminal data drawn from a different stationary equilibrium are not); and it aggregates using stationary-distribution identities unavailable along a transition.

Suppose, nonetheless, that aggregate consumption were antitone in the rate path—$C(t;r) \geq C(t;r')$ whenever $r \leq r'$ pointwise. Then the multiplier map would be antitone, and the standard two-sided iteration started from the edges of the band—each iterate an application of the natural computational scheme—would generate monotone sequences converging to a pair of paths $\underline{r}^* \leq \bar{r}^*$ bracketing *every* equilibrium, with uniqueness whenever the bracket collapses. The bracket would be computable, and its collapse checkable numerically. We emphasize that this monotonicity is a hypothesis: it holds in the stationary limit under a no-borrowing limit by Achdou et al. (2022a), and over short horizons up to $O(T)$ by the sensitivity structure of Section 5, but it is unproven for genuinely time-varying perturbations even when $\gamma \leq 1$.

For risk aversion above one, even the hypothesis is untenable as a general principle. Höfer (2025, Thm. 1.10) shows that for every $\gamma > 1$ there *exist* two-state economies (with a no-borrowing limit; the construction is a rare-disaster income process, and it is existential over economies, not universal) and rates $r_1 < r_2 < \rho$ at which stationary aggregate wealth satisfies $S_{\text{stat}}(r_1) > S_{\text{stat}}(r_2)$. Two caveats before drawing conclusions: the construction lives at the no-borrowing limit $\underline{a} = 0$, a boundary case of our maintained $\underline{a} < 0$ (where the hand-to-mouth channel $m_1\underline{a}$ vanishes identically), so its extension to $\underline{a} < 0$ rests on an unproven continuity argument; and the passage from stationary comparisons to transition paths is a turnpike heuristic. With those caveats, the picture such economies suggest is the following. On long horizons with data near a stationary point on the downward-sloping branch of S_{stat}, slowly varying path perturbations see a linearized clearing map with an approximate kernel (at a fold) or a wrong-signed diagonal: monotone methods and naïve continuation in the horizon would cross a singular linearization, the classical environment for bifurcation of self-fulfilling equilibrium paths—a believed future path of rates alters current consumption, reshapes the wealth distribution, and validates the belief. And when several stationary rates clear the same bond supply, data connecting two of them (initial distribution from one, terminal values from the other; both satisfy $S(0) = B$) admit, on long horizons, transition paths lingering near each steady state as natural candidates for distinct equilibria with identical endpoints.

None of this is proved, and we prefer to leave it that way in print: the honest summary is that short-horizon existence and uniqueness are established in Section 5 under explicit, verifiable hypotheses, long-horizon uniqueness for $\gamma \leq 1$ is a plausible conjecture waiting on a genuinely time-dependent monotonicity argument, and long-horizon uniqueness for $\gamma > 1$ is doubtful on the strength of the stationary counterexamples. For the empirically common calibrations with $\gamma > 1$, uniqueness of computed transition dynamics should be regarded as an assumption of computational practice—testable, for instance, by restarting path-iteration algorithms from dispersed initial guesses—rather than as a theorem.

8 Conclusion

The steady-state theory of heterogeneous-agent economies rests on two gifts: the interest rate summarizes the distribution in one number, and impatience pins households to the borrowing constraint, where their behavior is transparent. Along a transition both gifts are revoked. This paper's main theorem shows what replaces them: the operative equilibrium condition is the aggregate budget identity in flows—the form the computational algorithms already iterate on—and the equilibrium fixed point lives on a class of rate paths whose oscillations are self-consistently controlled by the one group that responds to the rate in real time, hand-to-mouth debtors. Existence and uniqueness over short horizons are established when that group's debt is smaller than the bond supply, $\bar{\theta} = |\underline{a}| \sup_t m_1/B < 1$: a measurable stability condition, load-bearing in the realistic boundary-layer regime. Around the theorem sit three findings worth carrying away: whether the constrained population survives to the end of a transition is decided by terminal conditions, not impatience; at zero net supply the constrained population *is* the market, and without it exact instant-by-instant clearing is the wrong demand; and for risk aversion above one, the uniqueness of long transitions is an open question on which the stationary evidence counsels doubt.

A The household problem: technical package

This appendix collects the assumptions and the analytical results behind Theorem 3.1 and Section 3.1. Throughout, the Hamiltonian is $H(p) = \sup_{c>0}\{u(c) - pc\} = \frac{\gamma}{1-\gamma} p^{1-1/\gamma}$ ($\gamma \neq 1$; $H(p) = -1 - \log p$ for $\gamma = 1$): finite only for $p > 0$ when $\gamma \leq 1$ (for $\gamma > 1$, $H(0) = 0$ but the maximizer escapes), strictly convex, with $H'(p) = -p^{-1/\gamma}$ singular as $p \downarrow 0$ for every γ.

A.1 Maintained assumptions

Assumption A.1 (primitives)**.** $\gamma > 0$; $\rho \geq 0$; $0 < y_1 < y_2$; $\lambda_1, \lambda_2 > 0$; $-\infty < \underline{a} < 0$; $T < \infty$; $B \in [0, \infty)$.

Assumption A.2 (price band)**.** $r \in \mathcal{K}_I$ with $I = [r_{\min}, r_{\max}]$, $r_{\max} < y_1/|\underline{a}|$; write $\bar{r} = \max\{|r_{\min}|, r_{\max}\}$ and $\underline{c} = y_1 + r_{\max}\underline{a} > 0$.

Assumption A.3 (terminal data)**.** $v_{T,j} \in C^1([\underline{a}, \infty))$, concave, strictly increasing, with constants $c_T^{\min} > 0$, $C_T > 0$ such that *(VT-u)* $\partial_a v_{T,j} \leq u'(c_T^{\min}) =: \bar{p}_T$ and *(VT-l)* $\partial_a v_{T,j}(a) \geq u'(C_T(1 + a - \underline{a}))$ for all a. Both hold for $v_T = v_\infty(\cdot\,; r^*)$ with $r^* \in I$, $r^* < \rho$, by the stationary binding equality and asymptotically linear stationary consumption.

Assumption A.4 (terminal layer regularity)**.** The terminal consumption policies $c_{T,j} = (u')^{-1}(\partial_a v_{T,j})$ satisfy the square-root modulus at the limit: on a neighborhood of $\underline{a}$,

$$|c_{T,j}(a) - c_{T,j}(\underline{a})| \ \leq \ C_{T,\text{lay}} \sqrt{a - \underline{a}}.$$

This is a genuine strengthening of Assumption A.3 (see Remark A.1); it holds for canonical terminal data $v_\infty(\cdot\,; r^*)$ by the stationary square-root law.

Assumption A.5 (initial data)**.** $G_0 = (G_{0,1}, G_{0,2})$ nonnegative measures on $[\underline{a}, \infty)$ with total mass one and finite first moment; compact support $[\underline{a}, \bar{A}]$ where stated; and no high-income mass at the limit, $G_{0,2}(\{\underline{a}\}) = 0$. Density regimes (D1)/(D1$'$) as in Section 2.2.

Assumption A.6 (the high-income type is unconstrained at the limit)**.** Along the band, $s_2(\underline{a}, t) > 0$ for all t: the high-income household at the borrowing limit strictly saves, so no type-2 boundary mass ever forms and the only atom is $m_1(t)$. Sufficient near the terminal date is $\partial_a v_{T,2}(\underline{a}) > u'(y_2 + r_{\max}\underline{a})$, which canonical terminal data satisfy (the high type saves at the constraint in the stationary equilibrium). This assumption is what licenses writing the contemporaneous clearing sensitivity as $B + |\underline{a}|m_1(t)$ in (5) and the anchoring function (6) with the type-1 atom only.

Remark A.1 (why Assumption A.4 is needed)*.* Fix $0 < \alpha < \frac{1}{2}$ and $c_0 > 0$, and set $c_T(a) = c_0 + (a - \underline{a})^\alpha$, $v_T(a) = v_T(\underline{a}) + \int_{\underline{a}}^a c_T(x)^{-\gamma} dx$. Then v_T is C^1, strictly increasing, concave, and satisfies (VT-u) with $c_T^{\min} = c_0$ and (VT-l) with a suitable C_T; if $c_0 < \min_{x \in I}(y_1 + x\underline{a})$ the terminal boundary policy is c_0. Yet $|c_T(a) - c_T(\underline{a})| = (a - \underline{a})^\alpha$ is not $O(\sqrt{a - \underline{a}})$. Hence the maintained bounds on terminal marginal values do not imply the square-root layer modulus, and any argument propagating layer estimates backward from T must assume it. (The square-root law of Lemma 3.4 generates the modulus on binding intervals *away* from the terminal layer, under its stated hypotheses—a C^1 path, uniform binding, $\nu_1 \geq \underline{\nu} > 0$—but not at $t = T$ from rough admissible terminal data.)

Remark A.2 (not assumed). Neither $\rho > 0$, nor $r(t) < \rho$, nor the stationary finiteness condition $\rho > (1-\gamma)r$ appears in this appendix. On a finite horizon the terminal valuation replaces transversality.

A.2 Value function and constrained viscosity solutions

The value function is $v_j(a,t) = \sup \mathbb{E}[\int_t^T e^{-\rho(s-t)} u(c_s) ds + e^{-\rho(T-t)} v_{T,y_T}(a_T)]$ over progressively measurable $c \geq 0$ maintaining $a_s \geq \underline{a}$; admissibility is nonempty uniformly over the band by Assumption A.2 (hold the state consuming $\underline{c}_{y_s}(s) > 0$). Standard arguments give pathwise wealth and budget bounds, finiteness and linear growth of v, concavity and strict monotonicity in a, and dynamic programming.

Definition A.3 (constrained viscosity solution)**.** (v_1, v_2), continuous on $[\underline{a}, \infty) \times [0,T]$, is a constrained viscosity solution if it is a viscosity subsolution of (1) on $(\underline{a}, \infty) \times [0,T)$, a supersolution on $[\underline{a}, \infty) \times [0,T)$ (boundary included), and matches the terminal data. For concave solutions the supersolution property at $\underline{a}$ is equivalent to (2) (Soner, 1986a; Capuzzo-Dolcetta and Lions, 1990).

Theorem A.4 (existence, comparison, uniqueness)**.** *Under Assumptions A.1–A.3, for every $r \in \mathcal{K}_I$ the value function is the unique constrained viscosity solution in the class of continuous functions concave, nondecreasing, of linear growth in a.*

Proof architecture. (i) Sub/supersolution properties from dynamic programming; the state-constraint supersolution property at $\underline{a}$ via Soner's inward cone, whose hypothesis is exactly $\underline{c}_j(t) \geq \underline{c} > 0$, uniform over the band (Soner, 1986a,b; Capuzzo-Dolcetta and Lions, 1990). (ii) The singular Hamiltonian is handled by truncation: comparison is proved for a globally Lipschitz $H_\flat$ agreeing with H on the compact gradient range delivered by Theorems A.6–A.7; solutions never see the truncation. (iii) Comparison for the weakly coupled system: the switching coupling is quasi-monotone, so the system comparison principles of Ishii and Koike (1991) and Engler and Lenhart (1991) combine with parabolic doubling (Crandall, Ishii, and Lions, 1992) and a linear penalization at infinity; the boundary is handled on the supersolution side by the inward cone, uniformly in t. (iv) Verification along Filippov solutions of the closed-loop dynamics (Appendix B). (v) C^1 regularity in a: a concave viscosity solution cannot kink where the Hamiltonian is strictly convex (Cannarsa and Sinestrari, 2004); switching terms are zeroth order and harmless. □

Remark A.5 (terminal boundary layer). The boundary inequality (2) holds at every $t < T$ even if the terminal data violate it at $t = T$: a marginal unit of wealth at the limit can always be consumed immediately at rate $\geq \underline{c}_j(t)$. If $\partial_a v_{T,j}(\underline{a}) < u'(\underline{c}_j(T))$, the boundary gradient jumps as $t \uparrow T$—a terminal boundary layer in $\partial_a v$, invisible in v.

A.3 Gradient bounds by barriers

Formally, $w_j = \partial_a v_j$ solves the cooperative transport system

$$\partial_t w_j + s_j\, \partial_a w_j = (\rho + \lambda_j - r(t))\, w_j - \lambda_j w_{-j}, \qquad w_j(\cdot, T) = v'_{T,j}; \tag{12}$$

the rigorous versions of the following run the barriers on vanishing-viscosity or implicit finite-difference approximations and pass to the limit.

Theorem A.6 (upper bound)**.** *For all j, a, t and $r \in \mathcal{K}_I$:*

$$\partial_a v_j(a,t) \;\le\; p_{\max} := \max\{\bar{p}_T,\, u'(\underline{c})\}\, e^{(r_{\max}-\rho)^+ T}.$$

Sketch. $W(t) = \max\{\bar{p}_T, u'(\underline{c})\}\exp(\int_t^T (r-\rho)^+)$ is a spatially constant supersolution of (12) (the required inequality $-(r-\rho)^+ W \le (\rho - r)W$ always holds); at the boundary the state-constraint mechanism can raise $w_j(\underline{a}, t)$ at most to the obstacle $u'(\underline{c}_j(t)) \le u'(\underline{c}) \le W(t)$; at T, $W \ge \bar{p}_T$. Cooperative comparison concludes. What replaced the stationary shortcut: stationarily $w_1(\underline{a}) = u'(y_1 + r\underline{a})$ exactly (binding under $r < \rho$); here binding may fail (Lemma 3.2), the bound comes from the obstacle or the terminal data, and the cap (VT-u) is a genuine assumption. □

Theorem A.7 (lower bound; linear consumption growth)**.** *There are $\kappa > 0$ and $C = C(\gamma, y_2, \bar{r}, \rho, \lambda)$ with $\partial_a v_j(a,t) \ge \kappa e^{-CT}(1+a-\underline{a})^{-\gamma}$, equivalently $c_j(a,t) \le \kappa^{-1/\gamma} e^{CT/\gamma}(1+a-\underline{a})$, uniformly over the band.*

Sketch. The ansatz $m(a,t) = \delta(t)(1 + a - \underline{a})^{-\gamma}$, identical across income states so the switching coupling cancels, is a subsolution of (12) for $\delta(t) = \kappa e^{-C(T-t)}$, using the one-sided drift bound $s_j \le y_2 + \bar{r}|\underline{a}| + r^+_{\max}(1 + a - \underline{a})$; at T it sits below $v'_{T,j}$ by (VT-l); at the boundary below the obstacle. □

A.4 Binding, dispersal, and the square-root law

Lemma 3.2 follows from the stability of v near T plus concavity: as $t \uparrow T$, $\partial_a v_1(\underline{a}^+, t) \to \max\{\partial_a v_{T,1}(\underline{a}), u'(\underline{c}_1(T))\}$, and comparing the implied desired consumption with hand-to-mouth income $\underline{c}_1(t)$ yields the two cases; Corollary 3.3 is the specialization to canonical terminal data.

Example A.8 (closed form: binding fails although $r < \rho$)**.** Remove income risk ($\lambda_1 = \lambda_2 = 0$, income y), let $r < \rho$ be constant, and take terminal data $v_T(a) = \omega_T^\gamma u(a+h_T)$ with $h_T > -\underline{a}$. The problem solves in closed form: $c(a,t) = (a + h(t))/\omega(t)$ with $h(t) = \int_t^T y\, e^{-r(s-t)} ds + h_T e^{-r(T-t)}$ (human wealth) and $\dot{\omega} = \frac{\rho - (1-\gamma) r}{\gamma}\omega - 1$, $\omega(T) = \omega_T$ (the inverse marginal propensity to consume). The boundary saving rate $s(\underline{a}, t) = y + r\underline{a} - (\underline{a} + h(t))/\omega(t)$ is strictly positive at $t = T$ for ω_T large (a patient, high-valuation terminal datum): binding fails on a whole terminal interval despite $r < \rho$. As $T - t \to \infty$, h and ω approach their stationary limits and binding is restored: the finite-horizon boundary behavior agrees with the stationary one only at distance $O(1)$ from T (a turnpike statement).

On Lemma 3.4. Matching orders in (12) at the boundary along a binding interval, with $w_1(\underline{a}, t) = u'(\underline{c}_1(t))$ so that $\partial_t w_1(\underline{a}, t) = u''(\underline{c}_1(t))\, \dot{r}(t)\underline{a}$, yields the displayed $\nu_1(t)$; the singular products cancel exactly as in the stationary case. The sign $\nu_1(t) \geq 0$ coincides with the complementarity (KKT) rate of the state-constraint multiplier, so the square-root law and binding degrade together. A rigorous version freezes coefficients on short time intervals and compares with sub- and super-boundary layers; we flag it as the delicate step of the package, and we emphasize again its hypotheses (C^1 path, binding, $\nu_1 \geq \underline{\nu} > 0$), which is why the corresponding aggregate estimates enter Corollary 5.4 as hypotheses verified on uniform binding regimes.

A.5 Stability in the rate path

Lemma A.9 (continuity and Lipschitz stability of the value)**.** *If $r_n \to r$ uniformly in $\mathcal{K}_I$, then $v^{r_n} \to v^r$ locally uniformly (half-relaxed limits plus comparison, the boundary obstacle converging uniformly by Assumption A.2), and the policies converge locally uniformly on $[\underline{a}, \infty) \times [0, T)$. Moreover*

$$|v_j^r(a,t) - v_j^{r'}(a,t)| \;\leq\; C^\star(1 + a - \underline{a})\, \|r - r'\|_{C([t,T])},$$

with $C^\star$ uniform over the band; only the future of the path enters.

Remark A.10 (policies across the boundary layer)*.* Pointwise Lipschitz stability of policies in the rate fails near the limit: the square-root layer's coefficient and its binding/release times move with r, making $\|c^r - c^{r'}\|_\infty \lesssim \|r - r'\|^{1/2}$ the honest generic rate. The equilibrium analysis needs less: in aggregates, the atom's consumption $y_1 + r(t)\underline{a}$ is exactly Lipschitz with constant $|\underline{a}|$, and the interior part meets a layer of width δ only on a set of mass $O(\delta)$ under (D1) and $O(\sqrt{\delta})$ under (D1′), whose contribution to aggregates is $O(\delta^{3/2})$ and $O(\delta)$ respectively; the pointwise loss integrates away, up to a logarithm under (D1′). This atom/interior division is the interface used in Appendix C.

B Distributional dynamics

Definition B.1 (measure-valued forward solutions)**.** Given policies c_j and drifts s_j, a narrowly continuous family $t \mapsto (G_{1,t}, G_{2,t})$ of nonnegative measures on $[\underline{a}, \infty)$ solves the forward equation if for all $\varphi_j \in C_b^1([\underline{a}, \infty) \times [0, T])$—including test functions *not* vanishing at $\underline{a}$—

$$\frac{d}{dt} \sum_j \int \varphi_j \, dG_{j,t} = \sum_j \int \Big[\partial_t \varphi_j + s_j \partial_a \varphi_j + \lambda_j(\varphi_{-j} - \varphi_j)\Big]\, dG_{j,t}, \qquad G_{j,0} = G_{0,j}.$$

This is the time-dependent version of the extended weak formulation of Achdou et al. (2022a) (Supplementary Appendix E), written out there for the two-type equation in stationary form with the time-dependent case declared analogous; unbounded test functions such as $\varphi = a$ are admitted by cutoff for solutions with uniformly bounded first moments.

Lemma B.2 (one-sided Lipschitz drift). *Since $c_j(\cdot, t)$ is nondecreasing, $(s_j(a,t) - s_j(b,t))(a - b) \le \bar{r}|a-b|^2$: the drift is one-sided Lipschitz uniformly in t and over the band, despite the square-root singularity (which has the contractive sign); and $s_1(\underline{a}, t) \ge 0$ always.*

Theorem B.3 (well-posedness of the forward equation). *There is a unique measure-valued solution: the law of the piecewise-deterministic process alternating the forward Filippov flows of $\dot{a} = s_j(a,t)$ with the Poisson switching clock. Trajectories merge at $\underline{a}$ (atom formation) but never cross; uniqueness follows by duality in the one-dimensional one-sided-Lipschitz transport theory (Filippov, 1960; Bouchut and James, 1998; Poupaud and Rascle, 1997), extended to the switching system by Duhamel. Compact supports propagate with explicit bounds, and the first moment $S(t)$ is Lipschitz in t uniformly over the band—the rigorous content of the flow identity in Lemma 4.2.*

Proposition B.4 (atom dynamics: growth and discontinuous release). *Let $m_1(t) = G_{1,t}(\{\underline{a}\})$. On binding intervals m_1 is absolutely continuous with*

$$\dot{m}_1(t) = \Phi_1(t) - \lambda_1 m_1(t), \qquad \Phi_1(t) = -\lim_{a \downarrow \underline{a}} s_1(a,t) g_1(a,t) \ \ge 0,$$

inflow through the square-root drift feeding the mass, leakage only through switching to high income. At a release time t_0 (binding fails just after t_0, e.g. the rate crosses below the terminal-implicit rate of Corollary 3.3), the mass detaches as a whole *and travels into the interior along the Filippov characteristic through $(\underline{a}, t_0)$, decaying at rate λ_1; hence m_1 drops to zero discontinuously, and re-accumulates if binding resumes. In particular $t \mapsto m_1(t)$ is of bounded variation but not continuous, and the contemporaneous clearing sensitivity $B + |\underline{a}| m_1(t)$ of Section 4 is switched on and off by the binding analysis of Section 3.1. On uniform binding regimes m_1 is absolutely continuous with derivative bounded uniformly over the band—the property consumed by hypothesis (M3).*

C Proofs and verification of the model hypotheses

Lemma 4.2. Take $\varphi_j = a$ in Definition B.1 (justified by cutoff): the switching terms cancel, leaving $\dot{S} = \sum_j \int s_j \, dG_{j,t} = Y + rS - C$. Given $S(0) = B$ and $C = Y + rB$, $\dot{S} = r(S - B)$, so $S \equiv B$ by Grönwall; conversely $S \equiv B$ forces $\dot{S} = 0$, i.e. $C = Y + rB$.

Proposition 4.3. (i) is Proposition 4.1. (ii): the value at dates $t \le t_0 - h$ responds to a perturbation supported on $(t_0 - h, t_0 + h)$ only through a Duhamel source active on a window of width $2h$ (Lemma A.9); the distribution responds directly only during the window (drift term $a\,\delta r$); integrating the policy response against the distribution over $[0, t_0]$ gives the bound, with the logarithmic factor under (D1′) inherited from Remark A.10. (iii): (ii) plus Arzelà–Ascoli; the first-kind character is the vanishing diagonal.

Verification of (M1)–(M4): status and arguments. We work on a symmetric sub-band $\mathcal{K}_R$ around r^{**} as in Theorem 5.3, on a *uniform binding regime*: the constraint binds along the sub-band throughout $[0, T]$ (sufficient conditions in Corollary 5.4), with Assumption A.4 in force. Write $\eta = \|r - r'\|_\infty$ and $\bar{m}_1 = \sup_{t, r\in\mathcal{K}_R} m_1(t; r)$. The household-level inputs are: the exact atom budget identity ($c_1(\underline{a}, t) = y_1 + r(t)\underline{a}$ under binding); the layer modulus $|c^r_j(a, t) - c^r_j(\underline{a}, t)| \le C_{\text{lay}}\sqrt{a - \underline{a}}$, *taken as an explicit package hypothesis uniformly over* $\mathcal{K}_R$—Lemma 3.4 derives it only for C^1, slowly varying paths with $\nu_1 \ge \underline{\nu} > 0$, and Assumption A.4 supplies it at the terminal layer, while the Schauder class X_T of Theorem 5.3 consists of paths with modulus $\Omega_T \asymp \sqrt{\cdot}$ that are not C^1, so the extension across $\mathcal{K}_R$ is assumed, not derived; interior policy stability at rate $L_1(T - t)\eta$ (differentiating the comparison argument behind Lemma A.9 where the value is locally $C^{1,1}$; the layer-crossing rate is the delicate step flagged in Remark A.10, used only in integrated form); the W_∞-flow stability $W_\infty(G^r_t, G^{r'}_t) \le L_2 t\eta$ (pathwise coupling, Lemma B.2, Grönwall); and the time-regularity of G and of m_1 (Theorem B.3, Proposition B.4).

(M1). As $T \to 0$, the value functions converge to the terminal data uniformly on compacts at rate $O(T)$, so interior policies converge to $c_{T,j}$ and the boundary policy to $\min\{c_{T,1}(\underline{a}), y_1 + r(t)\underline{a}\}$; distributions satisfy $W_\infty(G_t, G_0) \le \bar{s}T$ and $m_1(t) \to m_1(0)$ on binding regimes. Passing to the limit in $C(t; r)$ channel by channel gives $\sup_{r,t} |\Psi_T(r)(t) - A_0(r(t))| = \varepsilon_T \to 0$, with A_0 as in (6): the limit depends on the path only through the contemporaneous budget of the initially constrained.

(M2) and (M4). Decompose, for $r, r' \in \mathcal{K}_R$,

$$B\big[\Psi_T(r)(t) - \Psi_T(r')(t)\big] = \underbrace{\sum_j \int (c^r_j - c^{r'}_j)(\cdot, t)\, dG^{r'}_{j,t}}_{I} + \underbrace{\sum_j \int c^r_j(\cdot, t)\, d(G^r_{j,t} - G^{r'}_{j,t})}_{II}.$$

In I, the atom contributes $\le \bar{m}_1|\underline{a}|\eta$ (exact identity) and the interior $\le L_1 T\eta$. In II, couple the measures by the monotone quantile coupling at W_∞-distance $\hat{\eta} = L_2 T\eta$ and split the pairs: both at the atom (zero); mismatch pairs, one at the atom and one in the layer (consumption discrepancy $\le C_{\text{lay}}\sqrt{\hat{\eta}}$ by the layer modulus; layer mass $\le 2D\sqrt{\hat{\eta}}$ under (D1′); product $O(\hat{\eta})$—mass exchanged between atom and layer moves consumption only at second order because consumption is continuous across the limit); interior pairs (modulus $C_{\text{lay}}\hat{\eta}/\sqrt{x - \underline{a}}$ in the layer, Lipschitz outside), giving $\int_{\hat\eta}^{\delta_0}(\hat\eta/\sqrt{x})\,Dx^{-1/2}dx = D\hat\eta\log(\delta_0/\hat\eta)$. Summing:

$$\begin{aligned}
\|\Psi_T(r)-\Psi_T(r')\|_\infty &\le \bar\theta\eta + C_1T\eta + C_2T\eta\Big(1+\log\frac{D}{\eta}\Big) && \text{under (D1}'\text{)},\\
\|\Psi_T(r)-\Psi_T(r')\|_\infty &\le (\bar\theta + C_3T)\,\eta && \text{under (D1)},
\end{aligned}$$

which is (M4) in the two regimes (the $\log(D/\eta)$ form follows from $\log(\delta_0/\hat\eta)$ after absorbing constants, using $\eta\le D$ on $\mathcal{K}_R$) and, a fortiori, (M2).

(M3). Decompose $B[\Psi_T(r)(t)-\Psi_T(r)(s)] = [C(t;r)-C(s;r)]-[Y(t)-Y(s)]$. The atom's contribution to $C(t)-C(s)$ is

$$m_1(t)\big(y_1+r(t)\underline{a}\big)-m_1(s)\big(y_1+r(s)\underline{a}\big) = m_1(t)\,\underline{a}\,\big(r(t)-r(s)\big)+\big(m_1(t)-m_1(s)\big)\big(y_1+r(s)\underline{a}\big),$$

whose first term is bounded by $\bar m_1|\underline{a}|\,|r(t)-r(s)|$—the $\bar\theta|r(t)-r(s)|$ term of (8) with $\bar\theta = |\underline{a}|\bar m_1/B$—and whose second term is ω-type on uniform binding regimes, where m_1 is absolutely continuous with bounded derivative (Proposition B.4). The interior consumption and the distributional drift contribute moduli in $|t-s|$: the interior policies have a uniform time-modulus away from the terminal layer (bound $\partial_t v$ through the equation and the gradient bounds, then argue as for the layer modulus), and $W_\infty(G_t,G_s)\le\bar s|t-s|$ transfers through the layer/interior split above with $\hat\eta$ replaced by $\bar s|t-s|$. Y is Lipschitz. Collecting terms gives (8) with $\omega_T(h) = C(\sqrt{h}+h(1+\log_+(1/h)))$ under (D1$'$) (the square root from the layer's consumption modulus), a legitimate modulus.

Status. The contemporaneous ingredients (the $\bar\theta$-terms in (M3) and (M4), the θ_0-Lipschitz property of A_0) are exact. The smoothing ingredients rest on the household package under uniform binding plus Assumption A.4; within it, the layer-crossing policy stability, the rigorous square-root law, and the extension of the layer modulus from C^1 slowly varying paths to all of $\mathcal{K}_R$ (the preamble's explicit package hypothesis) retain the "delicate" flags recorded in Appendix A. Outside uniform binding regimes—dispersal episodes with their discontinuous m_1—(M3) fails as stated, and the analysis does not apply; this is a modeling boundary, not an oversight (cf. Section 5.3).

Proposition 6.1. Under binding on $[0,T]$, atom outflow is only through switching, giving $m_1(t)\ge m_1(0)e^{-\lambda_1 t}$; the map $\widetilde\Psi_0$ has denominator bounded below by $|\underline{a}|m_1(0)e^{-\lambda_1 T}>0$ and zero contemporaneous sensitivity (the solved-out diagonal), so Theorem 5.3 applies to it on a symmetric sub-band around its anchoring rate with $\bar\theta$ replaced by the (small) transmission constant of the mass channel; all constants scale by $1/(|\underline{a}|m_1(0))$, whence T^* degrading linearly in $m_1(0)$. Under (D1$'$), the separated atom mass is only $\frac{1}{2}$-Hölder stable in the path ($|m_1(t;r)-m_1(t;r')|\le C\sqrt{T\eta}$, the mismatch-mass computation above), which enters $\widetilde\Psi_0$ at order one: the (D1$'$) uniqueness question at $B=0$ remains open, as stated.

A remark on eliminating the diagonal at $B > 0$. The same solved-out map with denominator $B + m_1(t;r)|\underline{a}|$ removes the contemporaneous term altogether and, under (D1), yields the contraction conclusion without the hypothesis $\bar{\theta} < 1$. Under (D1′) it pays the $\frac{1}{2}$-Hölder price just described: aggregates are log-Lipschitz because atom–layer exchanges move consumption at second order, but any map that *separates* the atom is only $\sqrt{\cdot}$-stable. Under the realistic (D1′), the economic threshold $\bar{\theta} < 1$ is genuinely load-bearing.

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